

HKCEE 1984 Mathematics II

Label 1
1. $\frac{4}{(x-2)(x+1)} - \frac{3}{x^2-1}$

- A. $\frac{1}{(x-1)^2(x+1)}$
 B. $\frac{x+2}{(x-2)(x+1)(x-1)}$
 C. $\frac{x+10}{(x-2)(x+1)(x-1)}$
 D. $\frac{x-10}{(x-2)(x+1)(x-1)}$
 E. $\frac{x^2-3x-10}{(x-2)(x+1)(x-1)^2}$

Label 2
2. If $a = \frac{2b(2y-x)}{x-3y}$, then $y =$

- A. $\frac{a+2b}{3a+4b}x$
 B. $\frac{a-2b}{-3a+4b}x$
 C. $-\frac{a+2b}{3a+4b}x$
 D. $\frac{3a+4b}{a+2b}x$
 E. $\frac{-3a+4b}{a-2b}x$

Label 3
3. $(2^{n+1})^2 \times (2^{-2n-1}) \div 4^n =$

- A. 1
 B. 2^{2n-1}
 C. 2^{n^2+2n}
 D. 2^{n^2-2n}
 E. 2^{-2n+1}

Label 4
4. If $x+2$ is a factor of x^2+ax+b , then $2a-b+3 =$

- A. -7
 B. -1
 C. 0
 D. 1
 E. 7

Label 5
5. If α and β are the roots of $3x^2-x-1=0$, then $\frac{1}{\alpha^2} + \frac{1}{\beta^2} =$

- A. 7
 B. 3
 C. 1
 D. -1
 E. -5

Label 6
6. If $(\sqrt{3}-\sqrt{2})x=1$, then $x =$

- A. $\sqrt{3}+\sqrt{2}$
 B. $\frac{1}{\sqrt{3}+\sqrt{2}}$
 C. $\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{2}}$
 D. $\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{2}}$
 E. $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

Label 7
7. What is/are the root(s) of $\sqrt{5-x}=x-3$?

- A. 4 only
 B. 1 and 4 only
 C. -1 and -4 only
 D. -4 and 4 only
 E. -4, -1, 1 and 4

Label 8 The sum of the first ten terms of an arithmetic progression is 120. If the common difference is 4, then the first term is

- A. 12
- B. -6
- C. -2
- D. 2
- E. 6

Label 9 \$10 000 is invested for 2 years at 10% per annum, compounded half-yearly. The compound interest, correct to the nearest dollar, is

- A. \$12 155
- B. \$2155
- C. \$2100
- D. \$2000
- E. \$1025

Label 10 The equation $x^2 + kx + k = 0$ has equal roots (k being a constant). $k =$

- A. 4 only
- B. -4 only
- C. 0 or 4
- D. 0 or -4
- E. 4 or -4

Label 11 If $\frac{3x+2y}{x+5y} = 1$, then $\sqrt{x+y} : \sqrt{x-y} =$

- A. $\sqrt{5}$
- B. 3 : 2
- C. $\sqrt{5} : \sqrt{6}$
- D. $\sqrt{5} : 1$
- E. $\sqrt{7} : 2$

Label 12 A is 25% taller than B. B is 25% shorter than C. A's height : C's height =

- A. 1 : 1
- B. 5 : 4
- C. 3 : 4

- D. 5 : 3
- E. 15 : 16

Label 13 A rectangular box, without a lid, is 40 cm long, 30 cm wide and 10 cm height. The area of the external surface of the box is

- A. 2600 cm²
- B. 3400 cm²
- C. 3500 cm²
- D. 3800 cm²
- E. 12 000 cm²

Label 14 A man drives a car at 30 km/h for 3 hours and then at 40 km/h for 2 hours. His average speed for the whole journey is

- A. 14 km/h
- B. 30 km/h
- C. 34 km/h
- D. 35 km/h
- E. 70 km/h

Label 15 A alone can complete a job in 8 hours, B alone takes 12 hours and C alone takes 6 hours. After A and B have worked together on the job for 3 hours, C joins them. How much longer will they take to complete the job?

- A. 1 hour
- B. $1\frac{1}{2}$ hours
- C. 2 hours
- D. $2\frac{1}{2}$ hours
- E. 3 hours

Label 16 The marked price of a book is 20% above the cost price. If the book is sold at a discount of 10% off the marked price, what is the gain per cent based on the cost price?

- A. 8%
- B. 10%
- C. 12%

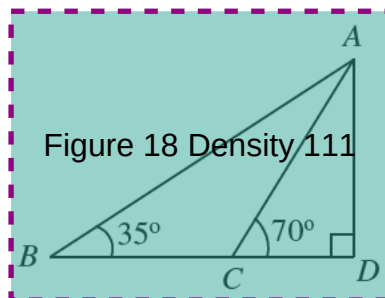
- D. 18%
- E. None of the above

Label 17

17. $\frac{\tan^2 \theta}{1 + \tan^2 \theta} + \cos^2 \theta =$

- A. 1
- B. $\frac{1}{2} + \cos^2 \theta$
- C. $\cos^2 \theta$
- D. $1 + \tan^2 \theta$
- E. $1 + \cos^2 \theta$

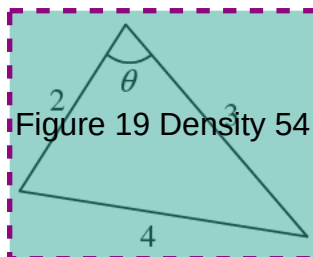
Label 18



In the figure, BCD is a straight line. $\angle ADC = 90^\circ$ and $BC = 10$. $AD =$

- A. $10 \cos 70^\circ$
- B. $10 \sin 70^\circ$
- C. $10 \tan 70^\circ$
- D. $\frac{10 \sin 20^\circ}{\sin 55^\circ}$
- E. $\frac{10 \tan 20^\circ}{\sin 55^\circ}$

Label 19

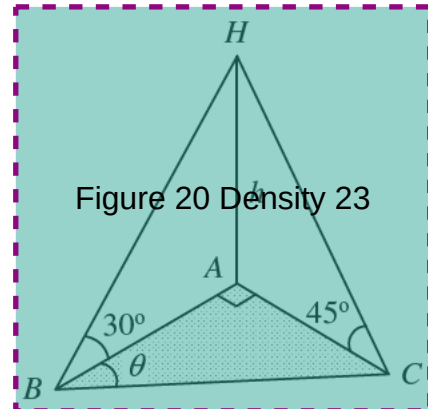


In the figure, $\cos \theta =$

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$

- C. $\frac{1}{4}$
- D. $\frac{1}{2}$
- E. $\frac{3}{4}$

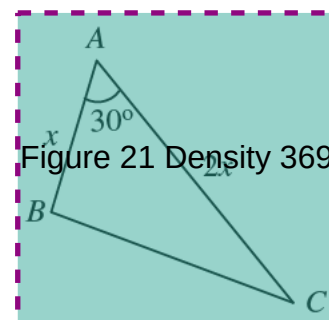
Label 20



In the figure, $\triangle ABC$ lies in a horizontal plane. $\angle BAC = 90^\circ$. HA is vertical and $HA = h$. $\tan \theta =$

- A. 1
- B. $\tan 30^\circ$
- C. $\frac{1}{\tan 30^\circ}$
- D. $h \tan 30^\circ$
- E. $\frac{h}{\tan 30^\circ}$

Label 21



In the figure, $AB = x$ and $AC = 2x$. The area of $\triangle ABC$ is 16. x (correct to 2 decimal places) is

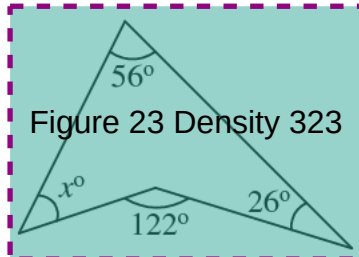
- A. 2.83
- B. 4.00
- C. 4.30
- D. 5.66

6.08

The sum of the interior angles of a convex polygon is greater than the sum of the exterior angles by 360° . How many sides has the polygon?

- A. 3
- B. 4
- C. 5
- D. 6
- E. 8

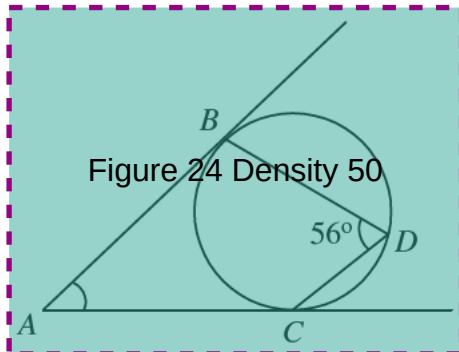
23



In the figure, $x = ?$

- A. 31
- B. 34
- C. 40
- D. 48
- E. It cannot be determined.

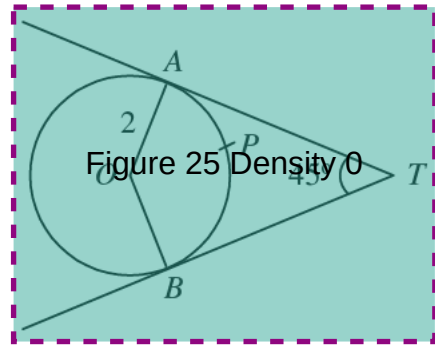
24



In the figure, AB and AC touch the circle at B and C respectively. $\angle A =$

- A. 30°
- B. 40°
- C. 50°
- D. 80°
- E. 85°

25



In the figure, O is the centre of the circle. TA and TB touch the circle at A and B respectively. OA = 2. The length of the arc APB is

- A. $\frac{\pi}{4}$
- B. $\frac{\pi}{2}$
- C. $\frac{3\pi}{4}$
- D. $\frac{3\pi}{2}$
- E. 3π

26

The point P divides AB internally so that $AP : PB = 2 : 1$. The coordinates of A and B are (x_1, y_1) and (x_2, y_2) respectively. The coordinates of P are

- A. $\left(\frac{2x_1 + x_2}{3}, \frac{2y_1 + y_2}{3} \right)$
- B. $\left(\frac{x_1 + 2x_2}{3}, \frac{y_1 + 2y_2}{3} \right)$
- C. $\left(\frac{2x_1 - x_2}{3}, \frac{2y_1 - y_2}{3} \right)$
- D. $\left(\frac{x_1 - 2x_2}{3}, \frac{y_1 - 2y_2}{3} \right)$
- E. $\left(\frac{x_1 + x_2}{3}, \frac{y_1 + y_2}{3} \right)$

27

The line $x + y + k = 0$ (k being a constant) passes through the centre of the circle $x^2 + y^2 - 2x + 4y - 6 = 0$. $k =$

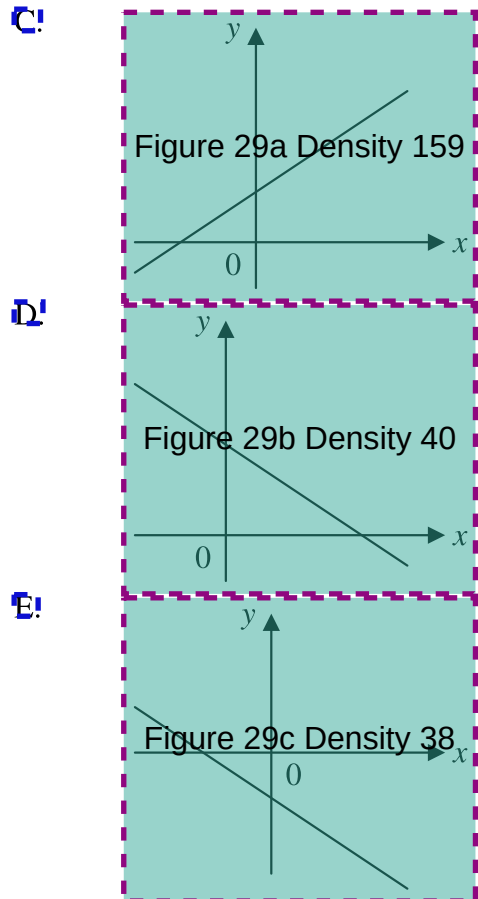
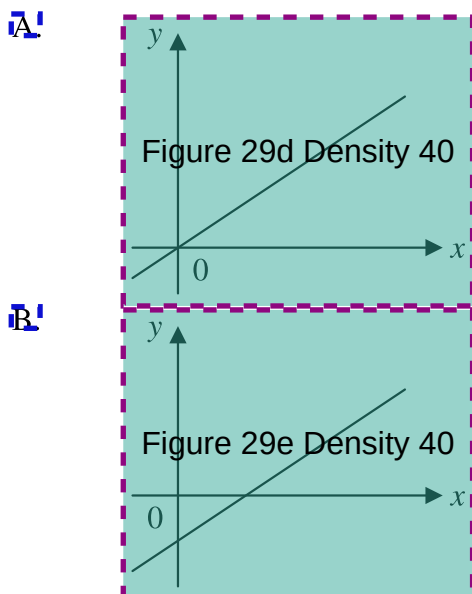
- A. -2
- B. -1
- C. 0
- D. 1
- E. 2

Label 28 The equation of a circle is $x^2 + y^2 - 2x + 5y - 7 = 0$.
28. Which of the following is/are true?

- I. The circle passes through the point $(-1, 1)$.
- II. The centre of the circle lies in the second quadrant.
- III. The circle intersects the x -axis at two points.

- A. II only
- B. III only
- C. I and II only
- D. II and III only
- E. I, II and III

Label 29 If a , b and c are positive real numbers,
29. which of the following graphs could represent the line $ax + by + c = 0$?



Label 30 The probability that John will win a
30. game is $\frac{1}{3}$ and the probability that he will lose is $\frac{2}{3}$. What is the probability that, in three games, he will win any two games and lose one game?

- A. $\frac{4}{27}$
- B. $\frac{2}{27}$
- C. $\frac{1}{27}$
- D. $\frac{2}{9}$
- E. $\frac{1}{9}$

Label 31 Two dice are thrown. What is the
31. probability of getting a sum of 8?

- A. $\frac{1}{12}$
 B. $\frac{1}{11}$
 C. $\frac{5}{36}$
 D. $\frac{1}{6}$
 E. $\frac{2}{9}$

Label 32 The standard deviation of the five numbers $a - 2d$, $a - d$, a , $a + d$, $a + 2d$ is

- A. 0.
 B. d .
 C. $\sqrt{2}d$.
 D. $\sqrt{5}d$.
 E. $\sqrt{10}d$.

Label 33 $x^2 - 9 \geq 0$ is equivalent to

- A. $x \geq \frac{3}{2}$ or $x \geq -\frac{3}{2}$.
 B. $\frac{3}{2} \leq x \leq -\frac{3}{2}$.
 C. $-\frac{3}{2} \leq x \leq \frac{3}{2}$.
 D. $x \geq -\frac{3}{2}$ or $x \leq \frac{3}{2}$.
 E. $x \leq -\frac{3}{2}$ or $x \geq \frac{3}{2}$.

Label 34 The graph of $y = x^2 + ax + b$ (a and b being constants) cuts the x -axis at $(2, 0)$ and $(h, 0)$, and cuts the y -axis at $(0, -2)$. $h =$

- A. -3
 B. -2
 C. -1
 D. 0
 E. 1

Label 35 If a and b are non-zero real numbers and $a > b$, which of the following must be true?

- I. $a^2 > b^2$
 II. $\frac{1}{a} > \frac{1}{b}$
 III. $a^3 > b^3$

- A. II only
 B. III only
 C. I and II only
 D. II and III only
 E. I and III only

Label 36 If $f(x) = (\log_{10} 2x) - x$, then $f(x+1) - f(x) =$

- A. $\log_{10} \frac{2-x}{x+1}$
 B. $\log_{10} \frac{x+1}{x}$
 C. $\log_{10} \frac{10(x+1)}{x}$
 D. $\log_{10} \frac{x+1}{10x}$
 E. $\log_{10} \frac{x+1}{x} - 2x$

Label 37 If $a \neq \pm 1$, then $1 + a^2 + a^4 + \dots + a^{2n} =$

- A. $\frac{1-a^{2n}}{1-a}$
 B. $\frac{1-a^{2n}}{1-a^2}$
 C. $\frac{1-a^{2n+1}}{1-a}$
 D. $\frac{1-a^{2n+1}}{1-a^2}$
 E. $\frac{1-a^{2n+2}}{1-a^2}$

Label 38 Which of the following must be geometric progression(s)?

- I. $\log_{10} 3, \log_{10} 9, \log_{10} 27, \log_{10} 81$

- II. 0.9, 0.99, 0.999, 0.9999
 III. 1, -3, 9, -27

- A. I only
 B. III only
 C. I and III only
 D. I and II only
 E. I, II and III only

Label 39
 39. a, b, c are positive numbers such that $\frac{a}{b} = \frac{b}{c} = k$ (k being a constant), which of the following must be true?

- I. $b^2 = k^2$
 II. $\frac{a+b}{b+c} = k$
 III. $\frac{a}{c} = k^2$

- A. II only
 B. III only
 C. I and II only
 D. II and III only
 E. I, II and III

Label 40
 40. Last year, a man saved 10% of his income. By how much per cent must his income be increased if his expenditure increased by 20% and he wants to save 20% of his income?

- A. 50%
 B. 35%
 C. 30%
 D. 20%
 E. 15%

84. The external and internal radii of a hollow metal sphere are 4cm and 3 cm respectively.

$$\frac{\text{Volume of metal}}{\text{Volume of the enclosed empty space}} =$$

- A. $\frac{1}{27}$
 B. $\frac{1}{3}$

- C. $\frac{4}{3}$
 D. $\frac{37}{27}$
 E. $\frac{64}{27}$

Label 42
 42. A solid metal sphere of volume 252 cm^3 is melted and recast into 3 smaller solid spheres whose radii are in the ratio 1 : 2 : 3. The volume of the smaller sphere is

- A. 5 cm^3
 B. 7 cm^3
 C. 14 cm^3
 D. 18 cm^3
 E. 28 cm^3

Label 43
 43. The base radii of two right circular cylinders are in the ratio 2 : 3. If the two cylinders have the same height, what is the ratio of their curved surface area?

- A. 2 : 3
 B. 4 : 9
 C. 8 : 27
 D. $\sqrt{8} : \sqrt{27}$
 E. None of the above

Label 44
 44. The greatest value of $\frac{3}{4+2\cos\theta}$ is

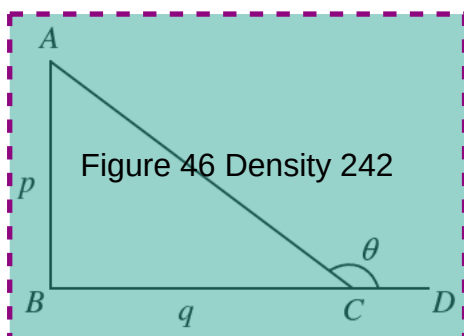
- A. 3
 B. $\frac{3}{2}$
 C. $\frac{3}{4}$
 D. $\frac{3}{5}$
 E. $\frac{1}{2}$

Label 45
45. If $0^\circ \leq \theta < 360^\circ$, the number of roots of the equation

$$2 \sin \theta + \frac{1}{\sin \theta} = 3$$

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

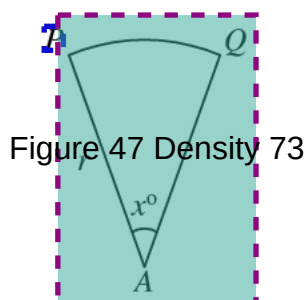
Label 46
46.



In the figure, $\angle B = 90^\circ$ and BCD is a straight line. If $AB = p$ and $BC = q$, then $\cos \theta =$

- A. $\frac{p}{q}$
- B. $\frac{p}{\sqrt{p^2 + q^2}}$
- C. $\frac{q}{\sqrt{p^2 + q^2}}$
- D. $\frac{-p}{\sqrt{p^2 + q^2}}$
- E. $\frac{-q}{\sqrt{p^2 + q^2}}$

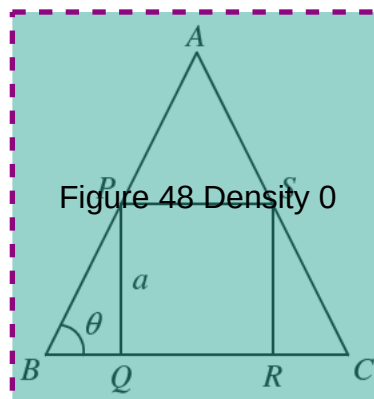
Label 47
47.



In the figure, the radius of the sector is r and $\angle POQ = x^\circ$. If the area of the sector is A , then $x =$

- A. $\frac{2A}{r^2}$
- B. $\frac{360A}{r^2}$
- C. $\frac{360A}{\pi r^2}$
- D. $\frac{180A}{r^2}$
- E. $\frac{180A}{\pi r^2}$

Label 48
48.



In the figure, $PQRS$ is a square inscribed in $\triangle ABC$. $AB = AC$ and $PQ = a$. $AB =$

- A. $a(\sin \theta + \frac{1}{2} \cos \theta)$
- B. $a(\sin \theta + \frac{1}{2} \sin \theta)$
- C. $a(\frac{1}{\sin \theta} + \frac{1}{2 \cos \theta})$
- D. $a(\frac{1}{\cos \theta} + \frac{1}{2 \sin \theta})$
- E. $\frac{2a}{\sin \theta}$

Label 49

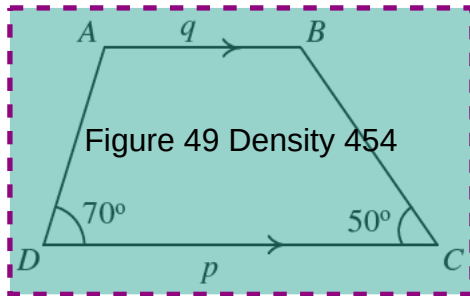


Figure 49 Density 454

In the figure, $AB \parallel DC$. $AB = q$ and $DC = p$. $BC =$

- A. $\frac{(p+q)\sin 50^\circ}{2\sin 70^\circ}$
- B. $\frac{(p+q)\sin 70^\circ}{2\sin 50^\circ}$
- C. $\frac{(p-q)\sin 70^\circ}{\sin 60^\circ}$
- D. $\frac{(p-q)\sin 70^\circ}{\sin 50^\circ}$
- E. $\frac{(p-q)\sin 50^\circ}{\sin 70^\circ}$

Label 50

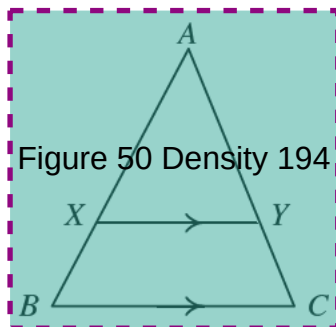


Figure 50 Density 194

In the figure, $XY \parallel BC$. $AX : XB = 2 : 1$. If the area of the trapezium $BCYX = 20$, then the area of $\triangle ABC =$

- A. 80
- B. 60
- C. 45
- D. 40
- E. 36

Label 51

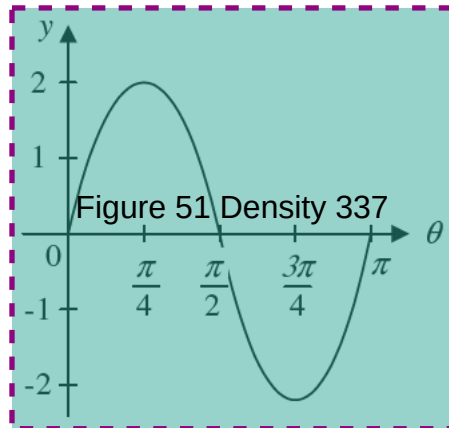


Figure 51 Density 337

The figure shows the graph of $y = a \sin k\theta$. What are the values of the constant a and k ?

- A. $a = 1$ and $k = 1$
- B. $a = 1$ and $k = 2$
- C. $a = 1$ and $k = \frac{1}{2}$
- D. $a = 2$ and $k = 2$
- E. $a = 2$ and $k = \frac{1}{2}$

Label 52

In $\triangle ABC$, $BC = a$, $AC = b$, $AB = c$ and $a > b > c$. Which of the following must be true?

- I. $\angle A > \angle B > \angle C$
- II. $b + c > a$
- III. $\angle B + \angle C > \angle A$

- A. I only
- B. II only
- C. III only
- D. I and II only
- E. II and III only

Label 53

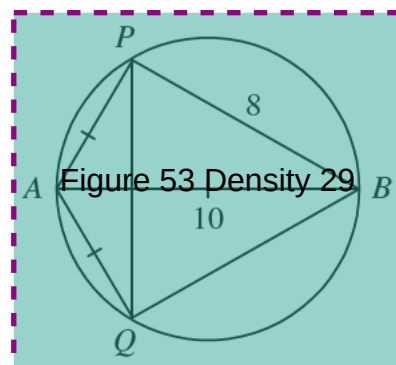
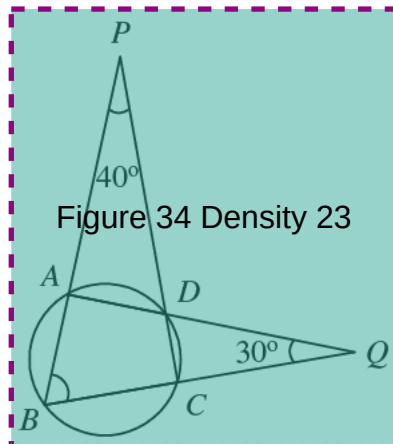


Figure 53 Density 29

In the figure, AB is a diameter of the circle. $AP = AQ$. $AB = 10$ and $BP = 8$.
 $PQ =$

- A. 5
- B. 6
- C. 6.4
- D. 8
- E. 9.6

Label 34



In the figure, the chords BA and CD , when produced, meet at P . The chords AD and BC , when produced, meet at Q .
 $\angle B =$

- A. 35°
- B. 40°
- C. 45°
- D. 50°
- E. 55°